Oliver Weißl

oliver.weissl@outlook.com • +436704063623 • oliverweissl.github.io • linkedin.com/in/oliver-weissl •
github.com/oliverweissl • Google Scholar • ORCID 0009-0008-7575-0187

Personal Profile

AI Researcher specializing in Deep Learning system testing and robotics, with formal training in Artificial Intelligence and Data Science and Industry experience in RAG and LLM/ GenAI solutions and deployment. Currently pursuing doctoral research on testing and validating deep learning systems through latent space analysis. Experienced in machine learning, Python, and evolutionary robotics, with research contributions spanning AI reliability, model evaluation, and autonomous systems.

Work Experience

NorCom Information Technology GmbH & CO KGaA September 2026 - now
Senior Data Scientist Munich, DE

- Developed and maintained a production predictive maintenance analytics application .
- Scaling RAG solutions from thousands to millions of files for agentic e-Discovery.
- Leading LLM testing, benchmarking, quality assurance, and AI governance for a large-scale public-sector AI infrastructure serving over 80,000 daily users.

fortiss GmbH September 2024 - June 2026
Scientific Researcher Munich, DE

- Consulted on RAG integration (construction SME, ∼200 employees).
- Consulted on the impact of GenAI on future software engineering.
- Developed a conversational RAG agent for research acceleration.
- Provisioned internal LLM infrastructure for small and large language models.
- Conducted applied AI research on AI adoption and deployment in SMEs and public-sector projects.
- Supervised Bachelor's and Master's projects on testing and validation of Deep Learning systems.

Vrije Universiteit Amstedam September 2023 - April 2024
Research Assistant Amsterdam, NL

- Developed and maintained *Revolve2*, a research framework for modular robot evolution.
- Conducted research on sim-to-real transfer by improving robot modeling and sensor-driven adaptation.
- Published research findings in leading venues within evolutionary robotics and AI.

Vrije Universiteit Amstedam April 2022 - June 2024
Teaching Assistant Amsterdam, NL

- Supported teaching in Machine Learning, Knowledge & Data, Computational Intelligence, and Human-Computer Interaction courses.
- Led practical sessions and tutorials with 50+ students, mentoring students and addressing technical questions.

Red Cross Austria October 2019 - June 2020
Paramedic Salzburg, AT

- Responsible for assessment of medical conditions in critical or emergencies
- Experienced in CPR, poly-trauma transportation, and immediate medical interventions

Sony DADC Europe Limited August 2017 & August 2018
IT Support Intern Salzburg, AT

Education

Technical University Munich September 2024 - Now
PhD in Generative Testing of DL Systems Munich, DE

- *Accepted Publications:* Targeted Deep Learning System Boundary Testing --- TOSEM
- *In Revision:* HyperNet-Adaptation for Diffusion-Based Test Case Generation --- TOSEM
- *Works in Progress:* Testing of VLMs & Neural Surrogates for CFD

Universiteit van Amsterdam September 2023 - July 2024
MSc in Information Studies: Data-Science Amsterdam, NL

- *Thesis:* Machine Learning Decoder for Topological Codes in 3 Dimensions

Vrije Universiteit Amsterdam September 2020 - July 2023
BSc in Artificial Intelligence Amsterdam, NL

- Part of the Honors program
- *Thesis:* Morphological-Novelty in Modular Robot Evolution

Leopold-Franzens Universität Innsbruck October 2022 - February 2023
Erasmus Semester in Computer Science Innsbruck, AT

Publications

Journals

[1] O. Weissl et al., "Targeted deep learning system boundary testing," *ACM Transactions on Software Engineering and Methodology*, 2025.

[2] J. Luo, K. Miras, C. Longhi, O. Weissl, and A. E. Eiben, "Lamarckian inheritance improves robot evolution in dynamic environments," *IEEE Transactions on Evolutionary Computation*, 2025.

Conference Proceedings

[3] O. Weissl and A. Eiben, "Morphological-novelty in modular robot evolution," in *2023 IEEE Symposium Series on Computational Intelligence (SSCI)*, 2023, pp. 1066–1071.

[4] O. Weissl and E. Egorov, "An equivariant machine learning decoder for 3d toric codes," in *2025 International Conference on Quantum Communications, Networking, and Computing (QCNC)*, 2025.

[5] J. Di Michele, O. Weissl, and A. Eiben, "Fertility during learning in evolutionary robot systems," in *Proceedings of the Genetic and Evolutionary Computation Conference*, 2025.

Conference Workshops

[6] G.-D. Kevin, O. Weissl, M. Karine, and V. K. Anna, "Interactive embodied evolution for socially adept artificial general creatures," in *Artificial life conference*, 2024.

[7] G. Merabishvili, O. Weißl, and A. Stocco, "Latent regularization in generative test input generation," *arXiv preprint arXiv:2602.15552*, 2026.

Preprints

[8] O. Weissl, V. Riccio, S. Kacianka, and A. Stocco, "Hypernet-adaptation for diffusion-based test case generation," *arXiv preprint arXiv:2601.15041*, 2026.

[9] X. Chen, O. Weissl, and A. Stocco, "Feature-aware test generation for deep learning models," *arXiv preprint arXiv:2601.14081*, 2026.

Skills

Programming	Python (Pandas, PyTorch, NumPy, TensorFlow, OpenCV, HuggingFace etc.), vLLM, Haystack, LangFuse, LangGraph, LiteLLM, Spark, LangChain, SQL
Miscellaneous	LaTeX, Git, Docker, Grafana, AWS, Azure Foundry.
General	GenAI, Deep Learning, Computer Vision, LLM, VLM, Robotics, Project Architecture

Languages

English	Professional proficiency
German	Native proficiency